Calendar Effects in Stock Markets: An Empirical Investigation

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Abstract

This study investigates the existence of calendar effects in financial markets, focusing on the January effect, the Halloween effect, the Turn-of-the-Month effect, and the Day-of-the-Week effect. Using daily stock return data, a regression framework is used to test for temporal cyclical patterns that cannot be explained within an efficient market framework. The results suggest that while some calendar effects are detectable, their magnitude and statistical significance are limited and have changed over time.

1 Introduction

Financial markets are often modelled as informationally efficient, in that asset prices quickly absorb new information through the actions of savvy arbitrageurs (as formalised by the Efficient Market Hypothesis (EMH) (Fama [1970])). Under this hypothesis, predictable patterns in returns based on public information should not persist, as they would be quickly exploited and eliminated.

However, empirical research has repeatedly documented the existence of calendar effects. Calendar effects are systematic and cyclical variations in returns exhibited over different periods of time, such as days of the week, months, or years within the trading calendar.

This report examines four well-known calendar anomalies:

- The January Effect
- The Turn-of-the-Month Effect
- The Halloween Effect
- The Day-of-the-Week Effect

Specifically, this report asks:

> *Do calendar effects have a statistically detectable impact on daily returns in the FTSE 100 and S&P 500 indices, and are any observed effects sensitive to alternative definitions and sample periods?*

The analysis considers two major equity markets: the FTSE 100 and the S&P 500. The FTSE 100 index tracks the London Stock Exchange, was established on 3rd January 1984, and represents the largest UK-listed companies. The S&P 500, established on 4th March 1957, is the main benchmark for large-cap US equities in the S&P U.S. index family. These indices therefore capture two of the largest and most liquid equity markets globally. Although they are similar in this respect, they differ in constituent companies, trading calendars, and broader institutional environments. Estimating the model separately for each index and then comparing them, therefore, allows an assessment of whether any detected calendar effects are market-specific or consistent across comparable markets. This is more informative than analysing a single market.

2 Literature Review

Fama [1970] first defined a market as efficient if prices "fully reflect" all available information, yet a plethora of empirical literature documents patterns at various frequencies that imply otherwise. The evidence is far from unanimous; many effects are supported in some samples but not others, and concerns have been raised repeatedly about data mining and publication bias ([Grebe and Schiereck, 2024, Agrawal and Tandon, 1994]). Schwert [2003] highlights an "attenuation after publication" phenomenon, wherein many anomalies appear to weaken or disappear in subsequent samples after publication, suggesting that either arbitrage or improved methodology erodes the effect over time. Perhaps such weakening is reminiscent of Goodhart's law (Goodhart [1984]) which states:

> *Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.*

At the annual level, the January effect remains one of the most studied anomalies. Early work by Rozeff and Kinney [1976] documents statistically significant variation in monthly returns, driven largely by unusually high January performance. Subsequent research identifies a correlation with firm size: both Keim [1983] and Ritter [1988] show that the January effect is concentrated among small-cap stocks, while value-weighted indices do not exhibit so much seasonality, implying that large firms contribute little to the effect. This result builds on earlier findings by Banz [1981], who first documented the impact of firm size in equity returns. The implication is that index-level studies may understate the true magnitude of the anomaly due to aggregation effects. There is also no consensus on this point; some literature disagrees that the effect is concentrated within smaller firms, as Clare et al. [1995] did for example.

Numerous explanations have been proposed for the January effect, although no single hypothesis fully accounts for the observed patterns. The most common is tax-loss selling (formalised by Roll [1983]), whereby investors sell underperforming stocks at the end of the year, followed by a rebound in January. To quote Roll, "There is downward pressure on stocks that have already declined during the year, because investors sell them to realize capital losses. After the year's end this price pressure is relieved and the returns during the next few days are large as those same stocks jump back up to their equilibrium values... we are obliged to test every theory, even one so patently absurd as this." While intuitive, evidence has been published to the contrary: Reinganum [1983] finds that the effect persists even when stocks most affected by tax-loss selling are excluded. Alternative hypotheses include omitted risk factors, seasonal variation in risk premiums, and portfolio rebalancing (also known as "window dressing"). Empirical tests of insider trading (e.g. Seyhun [1988]) or information-release hypotheses (e.g. Ritter [1988]) provide little supporting evidence, suggesting that the January effect likely reflects a combination of structural and behavioural factors rather than one overarching explanation.

At a semi-annual scale, the Halloween effect, colloquially called "Sell in May and Go Away", represents an especially interesting pattern of seasonality. Bouman and Jacobsen [2002] documents that returns are significantly higher during the November–April period than in May–October across a large international sample, with the effect present in 36 out of the 37 countries studied. The anomaly is robust to alternative time-window definitions and remains significant even after accounting for transaction costs.

Explanations for the Halloween effect are diverse but inconclusive. Early hypotheses focused on the possibility of driving sectors (particularly agriculture, real estate, and consumer goods industries), but studies found little correlation. Similarly, studies examining relationships with interest rates and trading volume found no great explanatory power. Behavioural and institutional explanations have also been proposed, including reduced trading activity during summer vacation periods and seasonal variation in investor attention. Some evidence suggests a relationship between vacations and the effect (proxied by measures such as population and market participation), but the lack of a reversed effect in the southern hemisphere challenges the idea ([Bouman and Jacobsen, 2002]). This trend has even been captured in mainstream media, with an article in the Evening Standard attributing the effect to "investors swanning off to enjoy Ascot... where they cannot find copies of their pink papers... leaving their nervous second-in-commands in charge" (May 1999). Still, the underlying drivers of the Halloween effect remain unclear.

Another effect is the Turn-of-the-Month effect. Lakonishok and Smidt [1988] uses long datasets to detect seasonal patterns, including higher returns around the turn of the month. McConnell and Xu [2008] also shows that the majority of equity returns occur within the last trading day of the month and the first few trading days of the next. In some samples, this window accounts for a disproportionate share of total returns, even finding that a strategy of trading only during the turn of the month would outperform the market.

Unlike many other calendar anomalies, Turn-of-the-Month effects are often attributed to institutional cash flows like salary payments, pension contributions, and investment behaviour that happen at the beginning/end of every month. Empirical support is mixed, however, as detailed by [McConnell and Xu, 2008]. Despite this, the Turn-of-the-Month effect is often regarded as one of the more persistent and meaningful forms of return seasonality.

At the shortest time scale, Day-of-the-Week effects (also known as weekday, weekend, Monday or Turn-of-the-Week effects) have also been widely documented, though the interpretation remains heavily contested. Early studies such as Cross [1973], French [1980], and Gibbons and Hess [1981] report systematically lower returns on Mondays relative to later days in the week. Subsequent work by Berument and Kiymaz [2001] shows the same, and additionally detects a peak in returns on Wednesdays.

A wide range of explanations have been proposed for weekday effects. Information-based hypotheses suggest that negative news is more likely to be released over weekends [Patell, 1976, Bouman and Jacobsen, 2002], while behavioural analyses highlight investor sentiment, including the so-called "Monday blues" [Rystrom and Benson, 1989]. Institutional explanations include settlement delays [Gibbons and Hess, 1981], and short-selling behaviour to avoid weekend risk [Chen and Singal, 2003]. Investigations that demonstrate persistence across trading cultures further complicate the interpretation; similar effects often occur on the first trading day of the week even in markets with different trading calendars, suggesting a more general "Turn-of-the-Week" phenomenon [Demirer and Karan, 2002].

More recent literature provides a more sceptical view of calendar effects. Meta-analytic investigations by Grebe and Schiereck indicate that such effects were strongest in earlier decades and have weakened in more recent samples. Results are very sensitive to methodology, sample period, and market, and some studies attribute earlier findings to data mining or publication bias [Harvey et al., 2016]. Some studies also suggest that effects are similarly exhibited across bonds, currencies, commodities, and cryptocurrencies, although the direction and magnitude vary widely Berument and Kiymaz [2001].

Overall, the literature suggests that calendar effects may exist across multiple temporal scales, but their persistence, magnitude, and interpretation remain contested. While early studies often report strong and stable anomalies, more recent evidence emphasises their sensitivity to methodology and of a potential lessening with time. As a result, any statistically significant calendar effects should be interpreted cautiously as observations could be linked to institutional trading patterns and investor psychology, rather than reliably exploitable opportunities. This perspective is consistent with the Adaptive Market Hypothesis [Lo, 2004], which views market efficiency as evolving and influenced by improvements and changing market conditions.

3 Data

3.1 Data Choice

The analysis uses daily index Close levels (S_t) of the FTSE 100 and of the S&P 500. A deliberate choice is made not to use dividend-adjusted indices, as they often are paid at the same times annually. This could skew the seasonality-based nature of the analysis.

3.1.1 Return Definitions

Daily returns are constructed as logarithmic returns and converted to basis points for readability:

$$r_t = 10000 \times (\ln(S_t) - \ln(S_{t-1})) \tag{1}$$

where S_t is the index Close level at time t. Log returns are used for their mathematical elegance and statistical convenience.

3.1.2 Data Sources

Market data is obtained from Yahoo Finance, which has data available for both the FTSE 100 and the S&P 500 for the duration of the sample period considered in this study.

3.1.3 Sample period

The baseline sample period spans 1990–2019, providing coverage across multiple market regimes and enabling subsample analysis.

3.2 Variable Construction

Dummy variables are defined as follows:

- January dummy ($D_t^{Jan} = 1$ if month is January).
- Other month dummies ($M_t^2, \ldots, M_t^{12}$), with one omitted category (January) to avoid multicollinearity.
- Halloween dummy ($D_t^{Hal} = 1$ if month $\in \{Nov, Dec, Jan, Feb, Mar, Apr\}$).

- Day-of-the-Week dummies ($D_t^{Tue}, \ldots, D_t^{Fri}$), with one omitted category (Monday) to avoid multicollinearity.
- Turn-of-the-Month dummy ($D_t^{TotM} = 1$ if last trading day or first three trading days).

Because some of these calendar dummies overlap, the coefficients should be treated with caution and interpreted relative to the other variables in the model.

4 Methodology

4.1 Model Specification

Calendar effects are tested using linear regression models of daily log returns.

The baseline specification contains the dummies most commonly used in literature:

$$r_t = \alpha + \beta_{Jan} D_t^{Jan} + \beta_{Hal} D_t^{Hal} + \sum_{d \in \{Tue, Wed, Thu, Fri\}} \delta_d D_t^d + \beta_{TotM} D_t^{TotM} + \varepsilon_t \tag{2}$$

With one weekday (Monday) omitted to avoid multicollinearity. An alternative specification includes dummies for every month:

$$r_t = \alpha + \sum_{m=2}^{12} \gamma_m M_t^m + \sum_{d \in \{Tue, Wed, Thu, Fri\}} \delta_d D_t^d + \beta_{TotM} D_t^{TotM} + \varepsilon_t \tag{3}$$

In this model, month indicators are defined for $m = 2, ..., 12$, with January ($m = 1$) similarly omitted as the reference category. Additionally, the January and Halloween dummies are omitted as they overlap entirely with multiple monthly definitions.

The baseline model allows direct tests of specific calendar anomalies, while the alternative specification provides a more flexible framework that captures general seasonal variation. In both models, coefficients are interpreted as average return differences relative to the omitted baseline category, conditional on the other variables included in the model.

4.2 Statistical Validation

4.2.1 Inference: HAC (Newey–West) Standard Errors

Daily return residuals are likely to exhibit short-run autocorrelation, impacting the validity of any detected statistical significance. To address this, standard errors are heteroskedasticity- and autocorrelation-consistent (HAC; Newey–West) standard errors.

A lag of five trading days is used as a baseline choice, reflecting one trading week.

4.3 Assumptions

The models use linear regression models estimated using ordinary least squares. There are several standard assumptions underlying this approach.

First, the model assumes a linear relationship between calendar regressors and expected returns. Since the objective is to estimate average return differences as opposed to model complex dynamics, this is sufficient.

Second, although HAC standard errors are used to make inference more robust to heteroscedasticity and autocorrelation, the time-varying nature of the underlying process is not directly modelled.

Third, the model assumes that all relevant seasonal structure is captured. Other factors, such as firm size, economic conditions, or risk premiums, are not explicitly modelled.

Finally, the analysis assumes that any detected calendar effects are stable over the sample period. This assumption is examined through subsample analysis, but variations within subsamples are still possible.

4.4 Implementation

1. **Download market data**
   Data frames of historical daily closing prices for the FTSE 100 and S&P 500 are retrieved for the chosen sample period, providing the raw time series used in the analysis.

2. **Preprocessing**
   Returns are converted into log returns using the standard transformation $\log(S_t/S_{t-1})$. These are scaled into basis points for easier interpretation.

3. **Build calendar dummies**
   Year, month, and day of the week are extracted from the date index, which are then used to define the calendar dummies.

4. **Split into subsamples**
   The full dataset is divided into pre-2008 and post-2008 periods.

5. **Define the regression design matrix**
   The relevant explanatory variables are compiled into a matrix of regressors, including a constant term to represent the intercept.

6. **Estimate the regression model using OLS**
   Ordinary Least Squares (OLS) is used to estimate the relationship between daily log returns and the calendar dummy variables, with HAC standard errors applied for inference.

7. **Visualisations**
   Plot the log returns against time for the baseline period and the average returns grouped by calendar month and day of the week.

8. **Estimate separate specifications**
   The baseline and monthly models are estimated separately for the FTSE 100 and S&P 500, and for each subsample period (eight total regressions).

9. **Output regression summaries for interpretation**
   Finally, print regression tables containing coefficients, standard errors, test statistics, p-values, and goodness-of-fit measures.

5 Results

Table 1: Baseline model regression results

	FTSE 100		S&P 500	
Variable	Pre-2008	Post-2008	Pre-2008	Post-2008
$Intercept$	-3.9136	-3.6185	3.8108	-5.6550
	(3.999)	(5.998)	(3.886)	(6.154)
D_{tue}	2.2638	8.4420	-4.0607	12.4710
	(5.159)	(7.196)	(5.090)	(8.313)
D_{wed}	-1.7056	-0.5692	-0.9491	4.2517
	(4.939)	(7.679)	(4.715)	(7.751)
D_{thu}	4.6112	-3.1612	-6.2165	7.7061
	(5.164)	(7.488)	(4.781)	(7.936)
D_{fri}	5.1024	3.0038	-6.4390	5.5913
	(4.745)	(7.231)	(4.664)	(7.129)
D_{totm}	11.7469***	3.1257	6.4787*	-0.4013
	(3.864)	(5.338)	(3.832)	(5.127)
D_{jan}	-7.3403	-10.3338	-2.2728	-8.5399
	(5.085)	(7.807)	(5.349)	(7.866)
D_{hal}	4.8018	5.8317	3.6766	5.9576
	(2.978)	(3.967)	(2.945)	(4.096)
Observations	4546	3031	4537	3020
R^2	0.003	0.002	0.002	0.002
Adj. R^2	0.002	-0.000	0.000	-0.001
F-statistic	2.425**	0.9004	1.107	0.7312
Prob(F-stat)	0.0177	0.505	0.356	0.646

HAC standard errors in parentheses (5 lags).

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

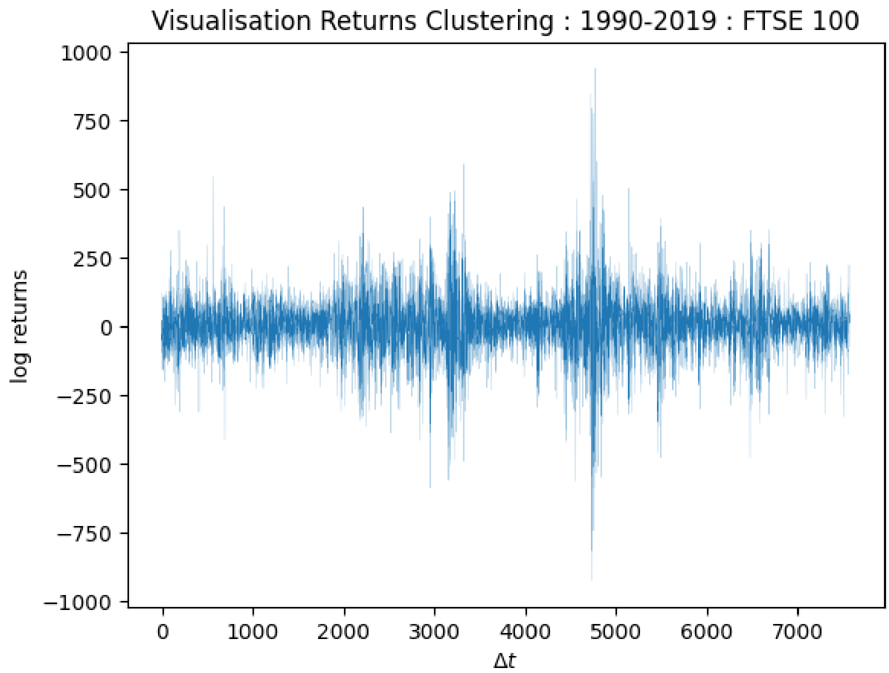

Figure 1: Daily FTSE 100 returns for the period 1990–2019, illustrating volatility clustering over time, motivating the use of HAC inference.

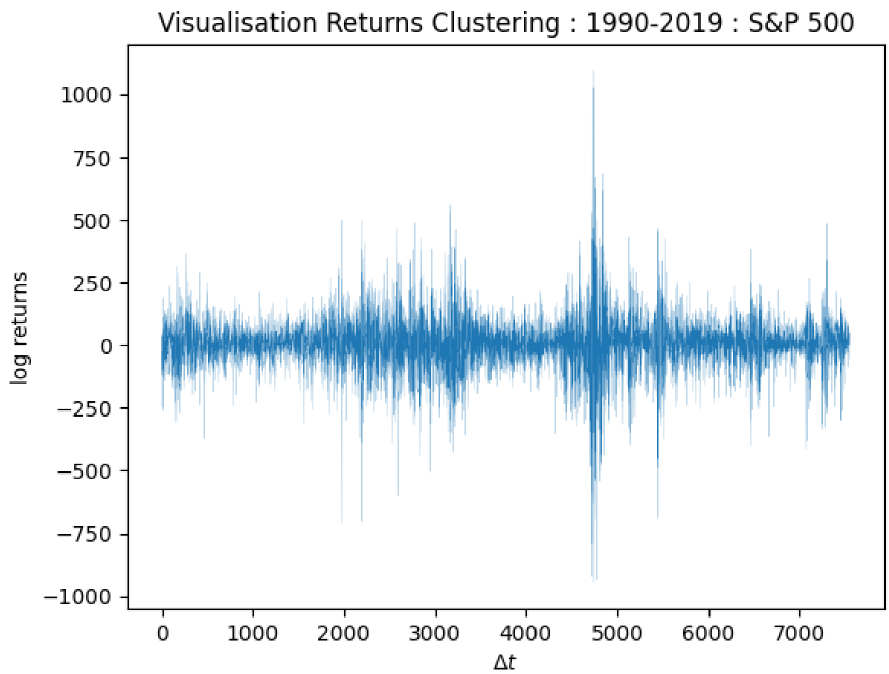

Figure 2: Daily S&P 500 returns for the period 1990–2019, illustrating volatility clustering over time, motivating the use of HAC inference.

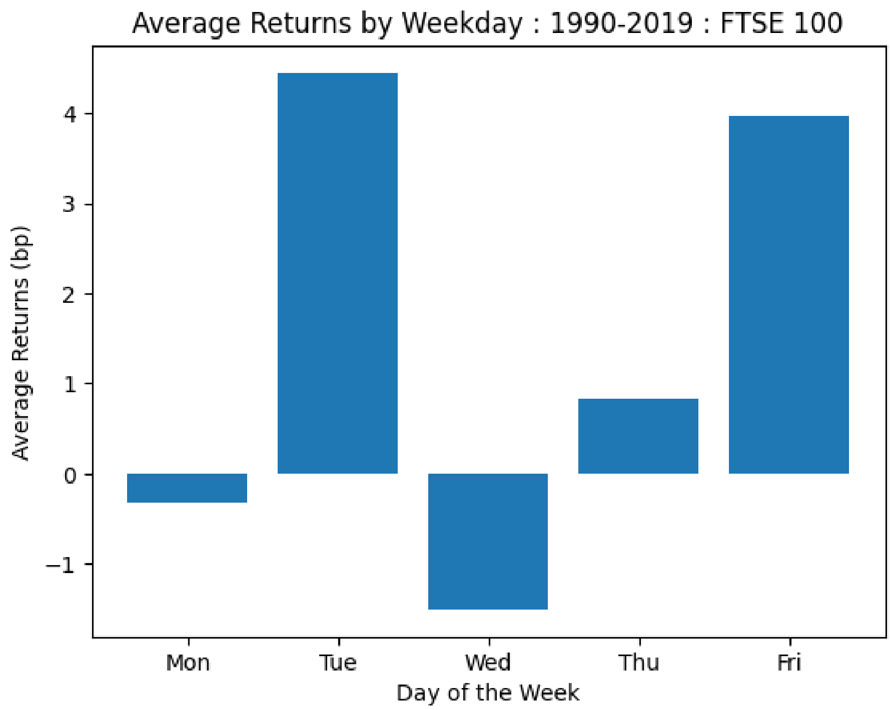

Figure 3: Mean FTSE 100 daily returns grouped by weekday for the period 1990–2019, measured in basis points.

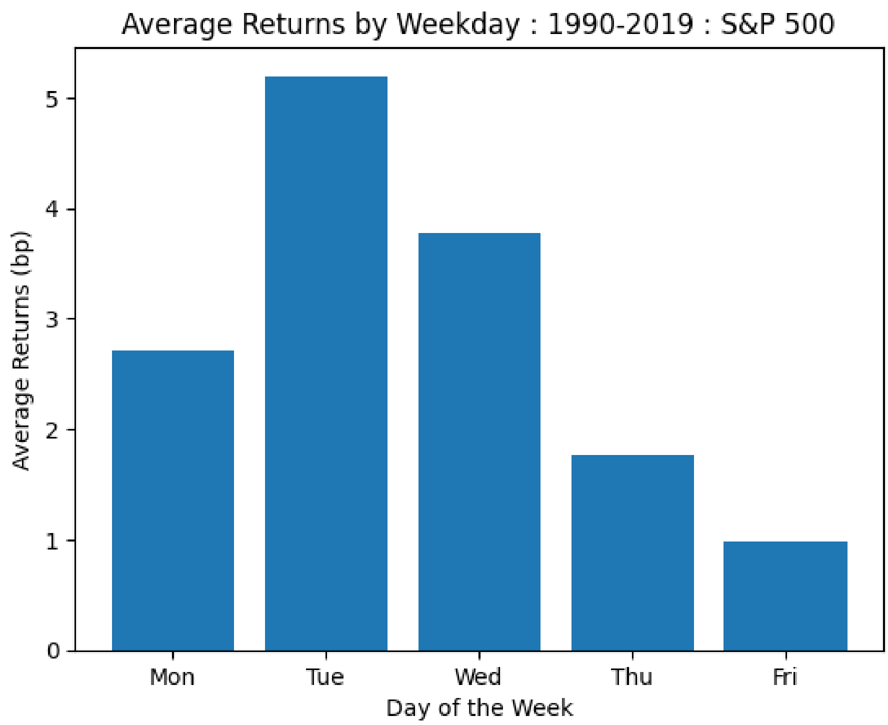

Figure 4: Mean S&P 500 daily returns grouped by weekday for the period 1990–2019, measured in basis points.

Table 2: Monthly model regression results

	FTSE 100		S&P 500	
Variable	Pre-2008	Post-2008	Pre-2008	Post-2008
$Intercept$	-6.4214	-8.1381	5.1274	-8.3642
	(5.788)	(9.066)	(5.846)	(9.057)
D_{tue}	2.2165	8.4284	-3.9973	12.5582
	(5.167)	(7.196)	(5.076)	(8.330)
D_{wed}	-1.7499	-0.5933	-0.8843	4.3646
	(4.938)	(7.683)	(4.712)	(7.773)
D_{thu}	4.5793	-3.1477	-6.0645	7.8358
	(5.176)	(7.498)	(4.795)	(7.959)
D_{fri}	5.0826	3.1217	-6.3570	5.8375
	(4.759)	(7.239)	(4.666)	(7.153)
D_{totm}	11.7382***	3.1190	6.5425*	-0.3759
	(3.867)	(5.303)	(3.817)	(5.101)
M_2	5.4785	9.8132	-2.8298	5.3362
	(6.076)	(9.973)	(6.721)	(10.584)
M_3	2.8363	5.8507	0.3966	11.5977
	(6.896)	(9.643)	(6.989)	(9.840)
M_4	9.5359	17.4395**	2.4885	12.5962
	(6.723)	(8.906)	(6.978)	(8.608)
M_5	3.8873	3.4385	5.0528	-0.7442
	(6.410)	(9.306)	(6.404)	(9.003)
M_6	-2.2483	-2.7464	-3.1513	-0.1988
	(6.174)	(9.471)	(6.222)	(9.314)
M_7	4.3128	15.8766*	-2.9679	12.9972
	(7.242)	(9.199)	(7.389)	(9.048)
M_8	0.9879	0.8431	-7.7735	-1.0940
	(7.003)	(10.544)	(7.407)	(9.875)
M_9	-3.9730	2.6853	-6.0600	1.9937
	(7.571)	(9.654)	(7.322)	(9.510)
M_{10}	11.8867	6.2001	6.2055	2.6852
	(7.469)	(11.627)	(7.545)	(12.415)
M_{11}	7.3328	3.9550	5.5131	8.0744
	(6.336)	(9.929)	(6.699)	(11.435)
M_{12}	11.9851*	15.3915	5.5054	4.7840
	(6.679)	(9.778)	(6.538)	(9.517)
Observations	4546	3031	4537	3020
R^2	0.005	0.004	0.004	0.003
Adj. R^2	0.002	-0.001	0.000	-0.003
F-statistic	1.531*	1.042	1.130	0.7395
Prob(F-stat)	0.0797	0.408	0.320	0.755

HAC standard errors in parentheses (5 lags).

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

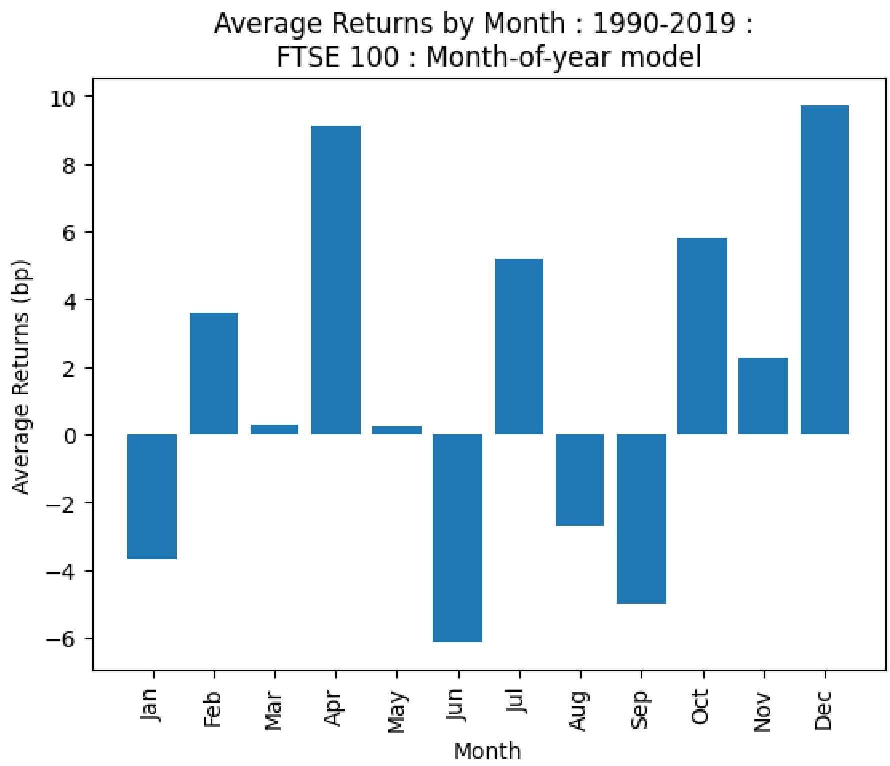

Figure 5: Mean FTSE 100 daily returns grouped by calendar month for the period 1990–2019, measured in basis points.

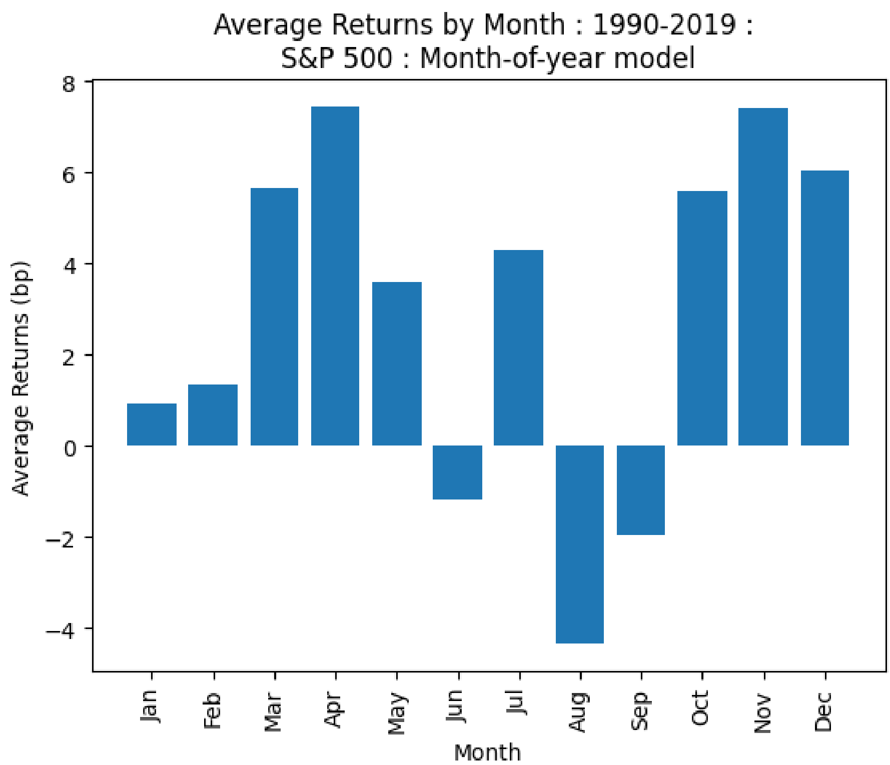

Figure 6: Mean S&P 500 daily returns grouped by calendar month for the period 1990–2019, measured in basis points.

6 Discussion

Overall, the results provide limited support for the existence of calendar effects in the FTSE 100 and S&P 500. In every specification, the models have little explanatory power for return variation, as shown by the low R^2 values. This means that any statistically significant effects should be interpreted with caution. The evidence points to a small number of weak and sample-dependent patterns.

The strongest result is the Turn-of-the-Month effect in the pre-2008 subsamples. For the FTSE 100, the Turn-of-the-Month dummy is positive and statistically significant at the 1% level in both the baseline and monthly models. For the S&P 500, the same coefficient is positive in the pre-2008 sample and shows borderline significance at the 10% level in both specifications. This is noteworthy because the Turn-of-the-Month effect is often regarded in the literature as one of the most persistent calendar anomalies. In this paper, it is the only effect that appears with some consistency across both the baseline and monthly specifications. However, even this result weakens substantially in the post-2008 period, where the coefficient becomes statistically insignificant in both markets. The implication is that the Turn-of-the-Month effect may have been present historically, but has weakened with time.

By contrast, the evidence for a January effect is weak. In all baseline specifications, the January dummy is statistically insignificant. Similarly, in the monthly models, the coefficients are not statistically different from the omitted baseline structure. This could be attributed to the commonplace view in literature that any January effect is undetectable in large-cap market indices unlike in small-cap stocks.

The Halloween effect also receives little support. Although the estimated Halloween coefficients are positive in every baseline regression, none are statistically significant and so are too weak to justify a strong claim that returns are systematically higher in the November–April period.

The monthly models provide some further descriptive value, but little firm statistical support for broad monthly seasonality. The monthly figures suggest that April and December tend to be relatively strong months for the FTSE 100, while April, November, and December appear relatively strong for the S&P 500. August and September look weaker in both markets. However, the formal regression results show that most monthly coefficients are not statistically distinguishable from the baseline. Only a few are marginally significant at the 10% level, such as December in the pre-2008 FTSE 100 sample and April or July in the post-2008 FTSE 100 sample. This may point to mild seasonal tendencies, but does not definitively confirm stable monthly effects.

A similar conclusion applies to the Day-of-the-Week effect. The weekday figures show Tuesday exhibiting the highest average return in both indices over the full 1990–2019 period, and Friday also appears relatively strong in the FTSE 100. This suggests that

Monday may generally be a weaker day. Regardless, none of the weekday dummies are statistically significant in either market in the regressions. So although the figures indicate that returns may differ across weekdays on average, the regression evidence does not support a robust Day-of-the-Week anomaly.

An important feature of the results is their instability over time. The pre-2008 period shows somewhat stronger evidence of temporal structure, particularly in the Turn-of-the-Month effect and the joint significance of some models (the F-statistic). In the post-2008 period, however, this evidence weakens greatly. Here, the models generally fail to show joint significance, and the stronger pre-2008 coefficients do not persist. This is arguably more important than any individual coefficient. If calendar anomalies were stable and intrinsic features of the market, they would survive more clearly across periods. Instead, the results suggest that any such effects are fragile and sensitive to the market environment.

This time variation is broadly consistent with the Efficient Market Hypothesis. The results do not suggest a market entirely without seasonal structure, nor do they point to strong, persistent, and easily exploitable anomalies. The Adaptive Market Hypothesis is a more forgiving interpretation, as it allows market efficiency to evolve over time as institutions, technology, arbitrage activity, and investor behaviour change. Under that interpretation, temporary calendar effects may exist in some periods and fade in others, as observed here.

The volatility plots show clear clustering in both indices. This visual evidence supports the use of HAC standard errors and emphasises the fact that daily returns are heteroskedastic. In such a context, it is difficult for simple linear dummy-variable models to explain much of the total variation in returns. The low explanatory power of the regressions is therefore not surprising.

Taken together, the findings show limited evidence for some effects, but not enough to support strong conclusions. There is some evidence for the Turn-of-the-Month effect especially in the earlier subsamples, but little evidence for January, Halloween, Day-of-the-Week, or broader monthly effects. The patterns that do appear are weak, unstable, and not obviously exploitable. For that reason, the results do not point to notable departures from market efficiency. Instead, they suggest that calendar effects in large-cap developed markets are limited and liable to weaken over time.

6.1 Limitations

Firstly, calendar effects are inherently vulnerable to sampling issues. Testing multiple dummies across markets, specifications, and subsamples increases the likelihood of false positives. Therefore, this paper places greater importance on the persistence of effects rather than only on statistical significance.

Furthermore, the analysis is conducted on large-cap indices. Prior literature suggests that some effects exist only in small-cap stocks. As such, the use of large-cap indices may be the reason for statistically insignificant effects or the lack of detection of effects altogether.

The empirical framework uses OLS estimation of log returns. While model inference is corrected using HAC standard errors, the model itself does not capture time-varying volatility. More sophisticated approaches, such as GARCH-in-mean models (as used by [Bollerslev et al., 1993] for example), were not used in favour of a simpler specification.

Although some variation in calendar dummies was tested, other definitions may be more informative. For example, Turn-of-the-Month windows vary across studies, and results may differ depending on these choices.

Lastly, extreme observations may disproportionately influence estimated coefficients, particularly during major market events such as the October 1987 crash or the 2008 financial crisis.

7 Conclusion

This study examined several well-known calendar effects in the FTSE 100 and S&P 500 using regression models with calendar dummy variables and HAC standard errors. The results suggest that calendar effects are not entirely absent, but they are limited in magnitude and inconsistent across specifications and sample periods. The strongest evidence is for a pre-2008 Turn-of-the-Month effect, particularly in the FTSE 100, while the January, Halloween, and Day-of-the-Week effects receive little convincing support in any period. The monthly models similarly provide only weak evidence of seasonal variation.

Overall, the findings suggest that calendar anomalies in large developed equity indices are limited at best and unstable with time. This is more consistent with a view of evolving market efficiency than with the idea of simple trading regularities. The results therefore show that calendar effects may occasionally be detectable in historical data, but they do not constitute compelling evidence of exploitable inefficiencies.

7.1 Future Work

Future work could explore several extensions:

- **Alternative return definitions:** Repeat regressions using simple returns instead of log returns.

- **Alternative dummy definitions:** Namely a different Turn-of-the-Month window (e.g. last day + first 1 day; last day + first 5 days).

- **Alternative asset classes:** Repeat the investigation on different asset classes.
- **Additional subsample splits:** Repeat the regression on different subsamples.
- **Volatility modelling:** Extend the model using GARCH-type specifications to examine whether results have any dependency on time-varying volatility.

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